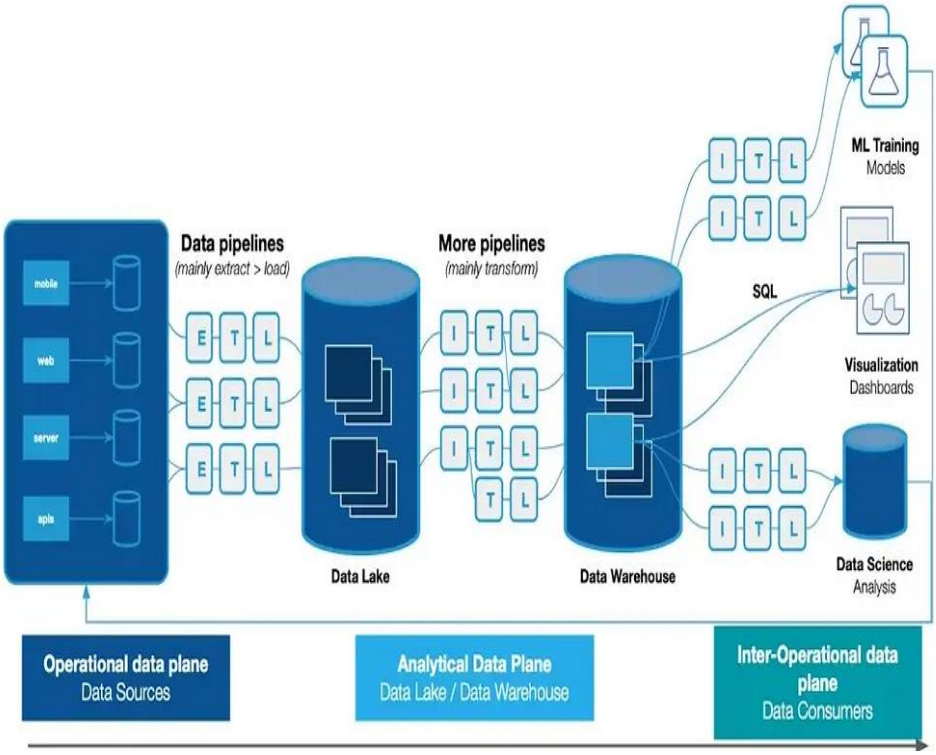
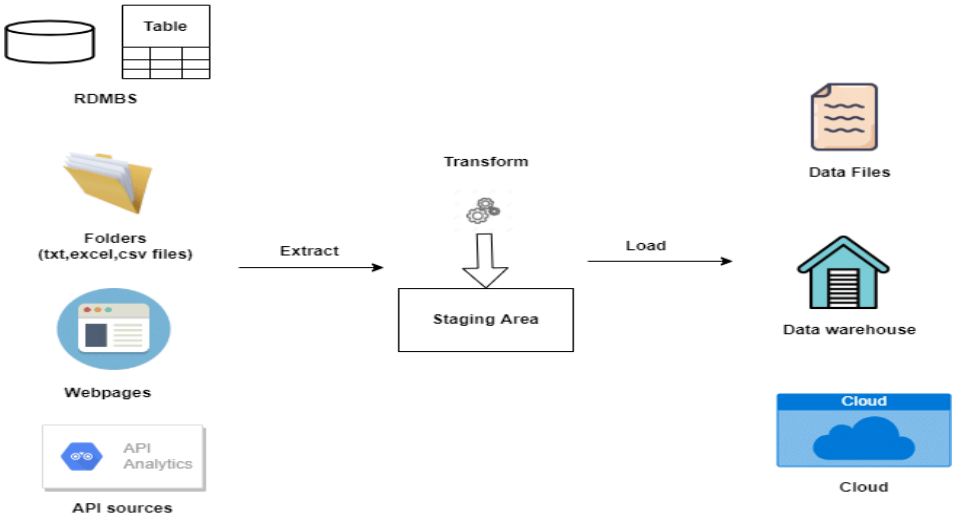
System Architecture

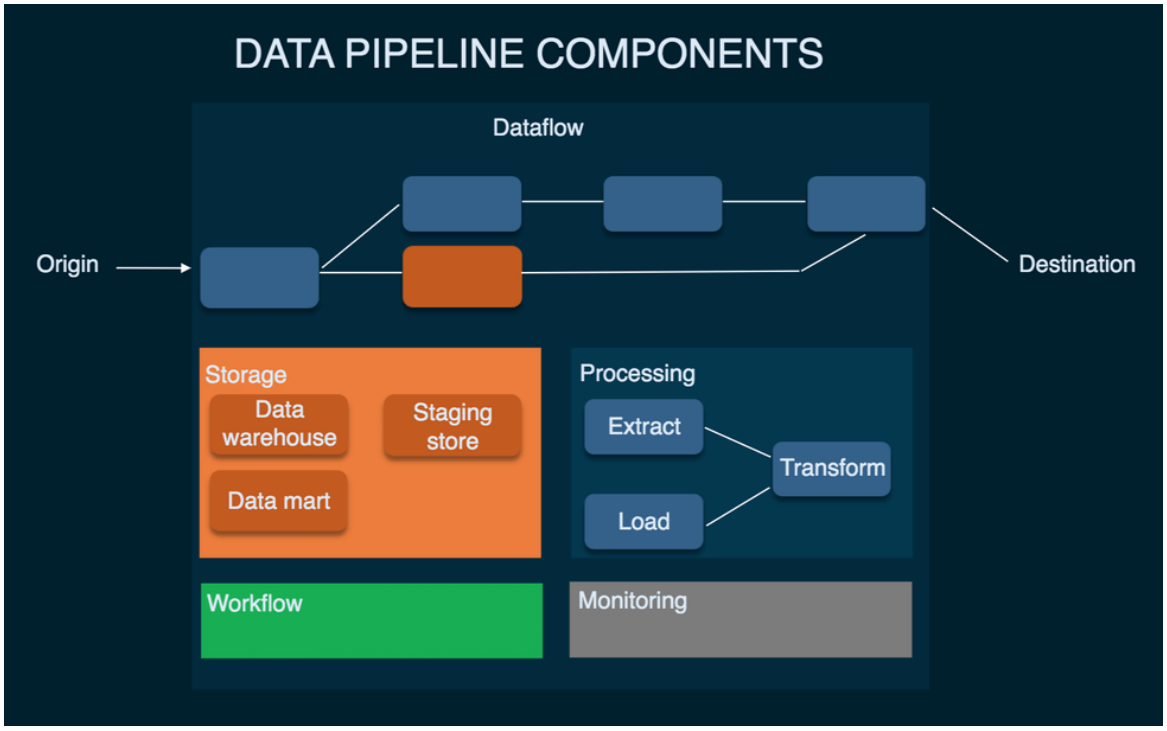
System Architecture Overview

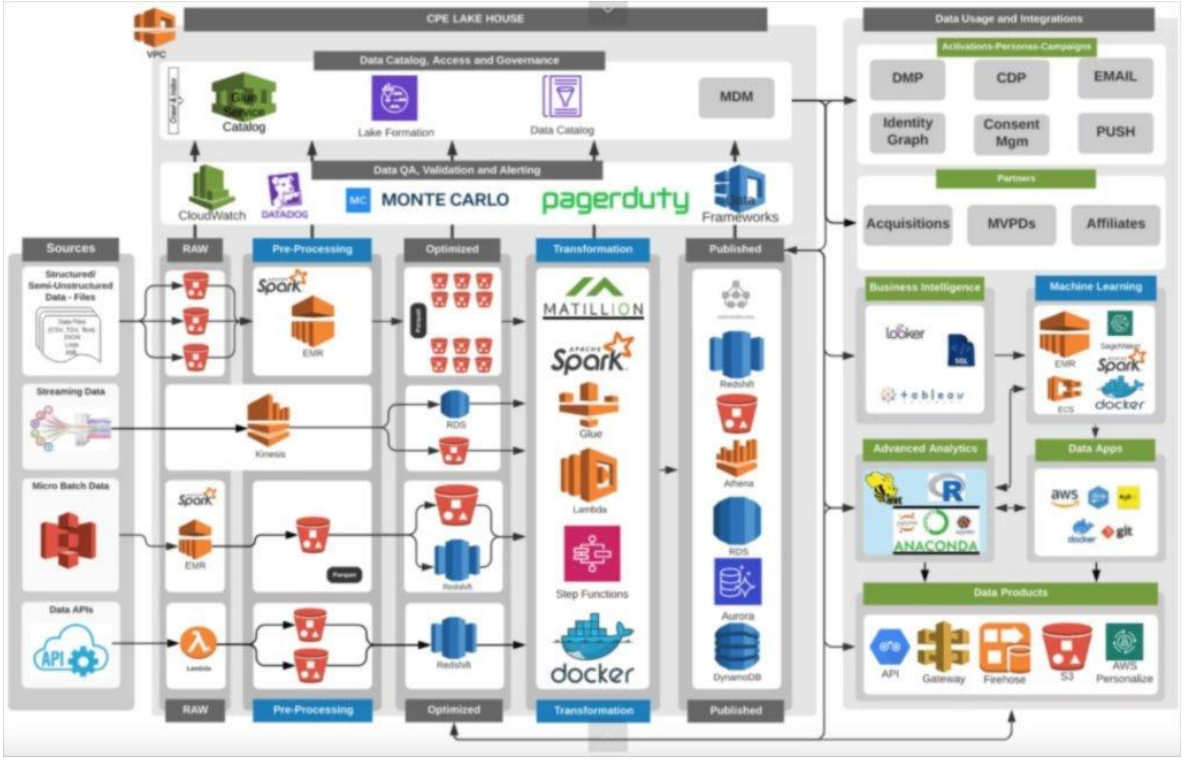
The system architecture represents the complete workflow of the End-to-End Data Engineering Pipeline. It shows how data flows from the source to the final output through different stages.



Architecture of ETL







Architecture Explanation

The pipeline starts with the **data source**, where raw data is collected in the form of CSV files. This data is then passed to the **ingestion layer**, where it is loaded into the system using Python.

Next, the data moves to the **processing layer**, where cleaning and transformation are performed using Pandas. This includes removing missing values and organizing the data into a structured format.

After processing, the data is stored in the **storage layer** as a structured dataset. This ensures that the data can be easily accessed and used for further analysis.

The stored data is then used by the **web application layer**, which is built using Streamlit. The application displays the processed data and basic analytical results such as total clicks.

Finally, the system produces the **output**, which includes structured data and visual insights that help in understanding the dataset.

Conclusion

This architecture demonstrates a complete data pipeline from data collection to visualization. It ensures smooth data flow, efficient processing, and clear output presentation. The system is simple, scalable, and suitable for demonstrating real-world data engineering concepts.